

SAIF UR REHMAN

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MSc Data Science and AI candidate, University of Liverpool. Research on geometry-aware curation of protein training data (dissertation), evidence use in feed-forward 4D reconstruction (first-author manuscript in development) and peer-reward misalignment in multi-agent LLMs (workshop submission, under review). Builds and evaluates ML systems in Python and PyTorch; applied AI intern at Rofsix, after systems, cloud and support engineering roles (2020–2025).

EDUCATION

MSc Data Science and Artificial Intelligence, University of Liverpool Sep 2025 – 2026
Dissertation: *Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction*, supervised by Dr Olga Anosova. Modules include Machine Learning and Bio-inspired Optimisation, Computational Intelligence and Applied AI.

BE Electrical (Telecommunication) Engineering, NUST, Pakistan 2016 – 2021

RESEARCH EXPERIENCE

Beyond AlphaFold / PDBClean: Geometry-Aware Curation of Protein Structure Training Data 2026
MSc dissertation, sole author • Python, PyTorch, OpenFold, MMseqs2, gemmi, Slurm github.com/saifrhman/702_BeyondAF

- Re-implemented and scaled the published Backbone Rigid Invariant (BRI) curation protocol as PDBClean, a configuration-driven pipeline over a fixed PDB snapshot (1 Jan 2026). A full re-run reproduced its output exactly.
- Implemented Elkin and Kurlin's (2023) compressed cover tree for exact L_∞ search, behind a length partition and a lossless lower-bound prefilter.
- Removed 78,754 geometric near-duplicates ($\leq 0.010 \text{ \AA}$) from 578,524 eligible chains, keeping 499,770. The cover tree's output matched an exhaustive check of all 3.53M candidate pairs, pair for pair.
- Found the geometric and sequence criteria nearly nested: 99.5% of geometric removals are sequence-identical to their representative (MMseqs2 on the same population).
- Prepared OpenFold inputs with fresh MSAs for a 142,056-chain, one-per-sequence set. From-scratch retraining under Slurm is in progress, with no accuracy results yet; the training scripts were adapted from a predecessor project.

Separating Solvers and Evaluators Is Not Enough: Peer-Reward Misalignment in Multi-Agent LLMs 2026
Co-author, four-person team • submitted to a NeurIPS 2026 workshop, under review

- Two LoRA policies on a shared frozen 7–9B base each solve a task and grade the other's answer. In most matched pairs, rewarding the peer's grade instead of the task metric lowered external task performance.
- Wrote the single-agent RL training loop (REINFORCE with a moving-average baseline) that the two-agent trainer builds on, and produced the single-agent replication figure.
- Follow-up: audited how policy-gradient credit reaches the grading tokens; designed a reward-topology comparison (objective reward, frozen judge, four-agent cycle, reciprocal pairs), not yet run.

Geometric Evidence Utilization in Feed-Forward 4D Reconstruction (working title) 2026 – present
First-author manuscript in development • research in progress • Target: CVPR 2027

- Asks whether feed-forward 4D reconstruction models use stronger geometric conditioning to estimate a physical quantity correctly, or fall back on a learned prior where the input does not determine it.
- Built a synthetic benchmark with an exact ambiguity endpoint (physically different worlds render byte-identical video) and a classical reference estimator. A ground-truth camera-pose intervention on Any4D and 4RC is in progress; no findings yet.
- Designing KINETIC, which represents dynamic scenes as independently transforming kinematic components with a separate global gauge. One transformation family and 20 test scenes are held out from all development.

Geographical Data and Computational Tools for Determining Restoration Priorities in Mangrove Ecosystems: A Scoping Review 2025 – 2026
Co-author, five-person team • manuscript in preparation • began as MSc group coursework

PRISMA-ScR review of 24 included studies. Restructured the manuscript and drafted large parts of the data-source, sensing-modality, methods and future-work sections.

EXPERIENCE

Applied AI Intern, Rofsix (UK) 2026 – present
Developing a real-time system that detects and redacts confidential information during screen sharing, combining OCR, transformer-based NER and image processing. Built and evaluated inference components with ONNX, OpenVINO and PaddleOCR, and profiled the pipeline for low-latency CPU execution.

Senior Systems Engineer, RoshTech Nov 2022 – Sep 2025
Built threat-detection and anomaly-analysis workflows over MongoDB, Graylog and the Elastic Stack to support incident investigation for enterprise clients.

Data Science Intern (part-time), Creative Tech Solutions (Islamabad) Jan 2024 – Apr 2024
Wrote Python scripts for data analysis and data cleaning for the analytics team.

Earlier roles

2020 – 2022

Senior Cloud Engineer, Tier3tech (Feb 2022 – Nov 2022) • Level 2 Support Team Lead, MiGo Innovations (Nov 2020 – Feb 2022)

SELECTED PROJECTS

F1 AI Copilot: Retrieval-Augmented QA over the FIA Regulations

github.com/saifrhman/f1-ai-copilot

Python • LangChain • Qdrant • FastAPI • Streamlit • pytest

- Question answering over the 2026 FIA F1 regulations (592 pages). A deterministic validator declines any answer whose citations, rule identifiers or figures are not in the passages it cites (sentences marked as inferences are exempt from the number check). The index fingerprints its build inputs; 1,442 tests run in CI.

Chest X-ray Classification (group coursework)

COMP534 Applied AI, University of Liverpool

PyTorch • EfficientNet-B3 • Swin V2-S • MaxViT-T • stratified 5-fold CV

- Team of four; 3,475 course-provided chest X-rays (Normal, Opacity, viral Pneumonia). Compared a custom 116,931-parameter CNN with EfficientNet-B3, Swin V2-S and MaxViT-T, each from scratch, frozen and fine-tuned.
- From scratch Swin V2-S failed to train (CV macro-F1 0.29); fine-tuned it scored highest (0.953) and reached test macro-F1 0.948, Cohen's κ 0.92 on 695 held-out images. Single run, not clinically validated; the Pneumonia class likely differs from the others in patient age and image source.

Hybrid Receipt Extraction: OCR, Rules and LayoutLMv3

github.com/saifrhman/noise-robust-ocr-pipeline

EasyOCR • LayoutLMv3 • PyTorch • Transformers • weak supervision

- EasyOCR, a rule parser and a LayoutLMv3 token classifier, merged field by field. SROIE labels only 4 fields, so rule-derived pseudo-labels for 13 more entity types are down-weighted (0.76–0.86 against 1.0) in the token loss. No evaluation results have been committed yet.

Football Intelligence Platform

github.com/saifrhman/football-analytics-platform

Python • BigQuery • dbt • FastAPI • Streamlit • GitHub Actions

- Pipeline from StatsBomb Open Data to BigQuery: bronze raw JSON, nine typed silver tables with explicit schemas, and dbt gold models (5 dimensions, 4 facts, 23 schema tests), served through FastAPI and Streamlit. Run end to end on a five-match sample; CI runs 39 unit tests, ruff and dbt parse.

TacticLens: Structural Retrieval over Football Tracking Data

github.com/saifrhman/tacticlens

Research design and foundation code • Python • Pydantic • FastAPI

- Research design (questions, baselines and rejection criteria fixed in advance) for retrieving passages of play with similar multi-player movement, and a tracking data model with seven typed observation statuses. The retrieval model, index and feedback loop are not yet built.

Sample-Efficient Learning: Active Learning and Bayesian Optimisation

scikit-learn • BoTorch • GPyTorch • active-learning-classification • closed-loop-hpo-botorch

- Entropy-based active learning reached 0.986 test accuracy after 60 labels, against 70 for random acquisition. GP Bayesian optimisation (Expected Improvement) reached best 5-fold CV accuracy 0.961 against 0.956 for random search in 33 random-forest tuning evaluations. Single seed each.

COMPETITIVE PROGRAMMING AND HACKATHONS

ICPC 2025: University of Liverpool team “Baby Harp Seal”: NWERC 2025, 107th; ICPC UK & Ireland Programming Contest 2025, 50th overall and 1st at the University of Liverpool site.

Hackathons: Liverpool Computer Society Hackathon, 1st place (team), deep-learning solution; Ultamation Hackathon, 2nd place, constraint-satisfaction scheduling.

SKILLS AND CERTIFICATIONS

Languages: Python, C++, SQL, Bash, R

LLMs and RL fine-tuning: LoRA/QLoRA, REINFORCE, multi-agent LLM training with peer-graded rewards, LLM-as-judge rewards, retrieval-augmented generation (LangChain, Qdrant), Hugging Face Transformers

ML training and evaluation: PyTorch, transfer learning (EfficientNet, Swin V2, MaxViT), Slurm GPU training and checkpointing (OpenFold), large-scale dataset curation and deduplication, cross-validation, error analysis, active learning, Bayesian optimisation (BoTorch, GPyTorch)

Geometry, vision and science: 3D/4D reconstruction models, protein structure data (OpenFold, MMseqs2, gemmi, Mol*), exact metric search (cover trees, k-d trees), LayoutLMv3, EasyOCR, PaddleOCR, OpenCV, YOLOv8, U-Net

Systems and data: HPC clusters (Slurm), OpenMP, ONNX, OpenVINO, FastAPI, Streamlit, BigQuery, dbt, PostgreSQL, MongoDB, Elastic Stack, Azure, Docker, Git, GitHub Actions, Linux

Certifications: Fundamentals of Accelerated Computing with Modern CUDA C++, NVIDIA (2026); Machine Learning Specialization, DeepLearning.AI on Coursera (2025); IBM Data Science Professional Certificate, IBM on Coursera (2025); Associate Data Engineer in SQL track, DataCamp (2025)